

Thermodynamics

Instructor

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Chapter 4 – Problems

**Problem numbers correspond to
5th Edition**

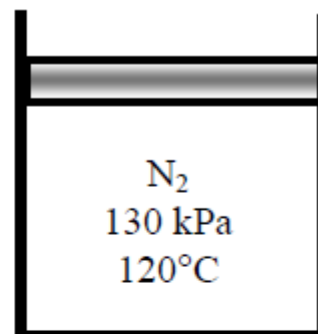
4-5 A piston–cylinder device initially contains 0.07 m^3 of nitrogen gas at 130 kPa and 120°C . The nitrogen is now expanded polytropically to a state of 100 kPa and 100°C . Determine the boundary work done during this process.

Properties The gas constant for nitrogen is $0.2968 \text{ kJ/kg}\cdot\text{K}$ (Table A-2).

Analysis The mass and volume of nitrogen at the initial state are

$$m = \frac{P_1 V_1}{RT_1} = \frac{(130 \text{ kPa})(0.07 \text{ m}^3)}{(0.2968 \text{ kJ/kg}\cdot\text{K})(120 + 273 \text{ K})} = 0.07802 \text{ kg}$$

$$V_2 = \frac{mRT_2}{P_2} = \frac{(0.07802 \text{ kg})(0.2968 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(100 + 273 \text{ K})}{100 \text{ kPa}} = 0.08637 \text{ m}^3$$



The polytropic index is determined from

$$P_1 V_1^n = P_2 V_2^n \longrightarrow (130 \text{ kPa})(0.07 \text{ m}^3)^n = (100 \text{ kPa})(0.08637 \text{ m}^3)^n \longrightarrow n = 1.249$$

The boundary work is determined from

$$W_b = \frac{P_2 V_2 - P_1 V_1}{1 - n} = \frac{(100 \text{ kPa})(0.08637 \text{ m}^3) - (130 \text{ kPa})(0.07 \text{ m}^3)}{1 - 1.249} = \mathbf{1.86 \text{ kJ}}$$

4-6 A piston–cylinder device with a set of stops initially contains 0.3 kg of steam at 1.0 MPa and 400°C. The location of the stops corresponds to 60 percent of the initial volume. Now the steam is cooled. Determine the compression work if the final state is (a) 1.0 MPa and 250°C and (b) 500 kPa. (c) Also determine the temperature at the final state in part (b).

Analysis (a) The specific volumes for the initial and final states are (Table A-6)

$$\left. \begin{array}{l} P_1 = 1 \text{ MPa} \\ T_1 = 400^\circ\text{C} \end{array} \right\} \nu_1 = 0.30661 \text{ m}^3/\text{kg} \quad \left. \begin{array}{l} P_2 = 1 \text{ MPa} \\ T_2 = 250^\circ\text{C} \end{array} \right\} \nu_2 = 0.23275 \text{ m}^3/\text{kg}$$

Noting that pressure is constant during the process, the boundary work is determined from

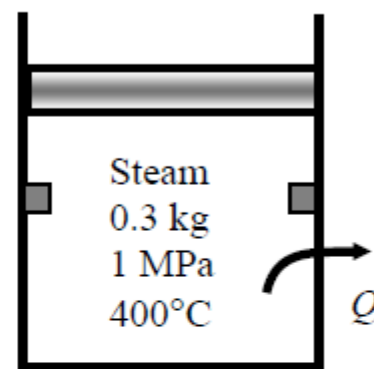
$$W_b = mP(\nu_1 - \nu_2) = (0.3 \text{ kg})(1000 \text{ kPa})(0.30661 - 0.23275) \text{ m}^3/\text{kg} = \mathbf{22.16 \text{ kJ}}$$

(b) The volume of the cylinder at the final state is 60% of initial volume. Then, the boundary work becomes

$$W_b = mP(\nu_1 - 0.60\nu_1) = (0.3 \text{ kg})(1000 \text{ kPa})(0.30661 - 0.60 \times 0.30661) \text{ m}^3/\text{kg} = \mathbf{36.79 \text{ kJ}}$$

The temperature at the final state is

$$\left. \begin{array}{l} P_2 = 0.5 \text{ MPa} \\ \nu_2 = (0.60 \times 0.30661) \text{ m}^3/\text{kg} \end{array} \right\} T_2 = \mathbf{151.8^\circ\text{C}} \quad (\text{Table A-5})$$



4-7 A piston–cylinder device initially contains 0.07 m^3 of nitrogen gas at 130 kPa and 120°C . The nitrogen is now expanded to a pressure of 100 kPa polytropically with a polytropic exponent whose value is equal to the specific heat ratio (called *isentropic expansion*). Determine the final temperature and the boundary work done during this process.

Properties The properties of nitrogen are $R = 0.2968 \text{ kJ/kg}\cdot\text{K}$, $k = 1.4$ (Table A-2a)

Analysis The mass and the final volume of nitrogen are

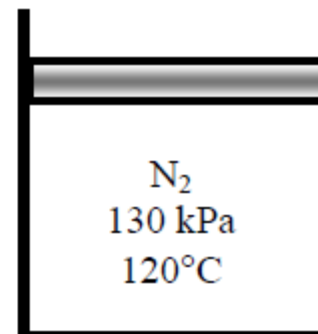
$$m = \frac{P_1 V_1}{RT_1} = \frac{(130 \text{ kPa})(0.07 \text{ m}^3)}{(0.2968 \text{ kJ/kg}\cdot\text{K})(120 + 273 \text{ K})} = 0.07802 \text{ kg}$$

$$P_1 V_1^k = P_2 V_2^k \longrightarrow (130 \text{ kPa})(0.07 \text{ m}^3)^{1.4} = (100 \text{ kPa})V_2^{1.4} \longrightarrow V_2 = 0.08443 \text{ m}^3$$

The final temperature and the boundary work are determined as

$$T_2 = \frac{P_2 V_2}{mR} = \frac{(100 \text{ kPa})(0.08443 \text{ m}^3)}{(0.07802 \text{ kg})(0.2968 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})} = \mathbf{364.6 \text{ K}}$$

$$W_b = \frac{P_2 V_2 - P_1 V_1}{1 - k} = \frac{(100 \text{ kPa})(0.08443 \text{ m}^3) - (130 \text{ kPa})(0.07 \text{ m}^3)}{1 - 1.4} = \mathbf{1.64 \text{ kJ}}$$



4-8 A mass of 5 kg of saturated water vapor at 300 kPa is heated at constant pressure until the temperature reaches 200°C. Calculate the work done by the steam during this process. *Answer: 165.9 kJ*

Assumptions The process is quasi-equilibrium.

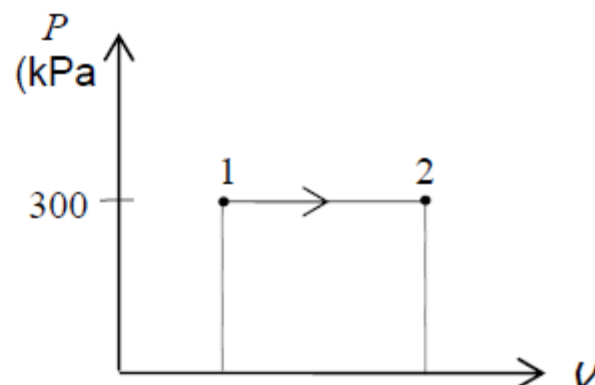
Properties Noting that the pressure remains constant during this process, the specific volumes at the initial and the final states are (Table A-4 through A-6)

$$\left. \begin{array}{l} P_1 = 300 \text{ kPa} \\ \text{Sat. vapor} \end{array} \right\} \nu_1 = \nu_g @ 300 \text{ kPa} = 0.60582 \text{ m}^3/\text{kg}$$

$$\left. \begin{array}{l} P_2 = 300 \text{ kPa} \\ T_2 = 200^\circ\text{C} \end{array} \right\} \nu_2 = 0.71643 \text{ m}^3/\text{kg}$$

Analysis The boundary work is determined from its definition to be

$$\begin{aligned} W_{b,\text{out}} &= \int_1^2 P dV = P(V_2 - V_1) = mP(\nu_2 - \nu_1) \\ &= (5 \text{ kg})(300 \text{ kPa})(0.71643 - 0.60582) \text{ m}^3/\text{kg} \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= \mathbf{165.9 \text{ kJ}} \end{aligned}$$



Discussion The positive sign indicates that work is done by the system (work output).

4-9 A frictionless piston–cylinder device initially contains 200 L of saturated liquid refrigerant-134a. The piston is free to move, and its mass is such that it maintains a pressure of 900 kPa on the refrigerant. The refrigerant is now heated until its temperature rises to 70°C. Calculate the work done during this process. *Answer: 5571 kJ*

Assumptions The process is quasi-equilibrium.

Properties Noting that the pressure remains constant during this process, the specific volumes at the initial and the final states are (Table A-11 through A-13)

$$\left. \begin{array}{l} P_1 = 900 \text{ kPa} \\ \text{Sat. liquid} \end{array} \right\} \nu_1 = \nu_f @ 900 \text{ kPa} = 0.0008580 \text{ m}^3/\text{kg}$$

$$\left. \begin{array}{l} P_2 = 900 \text{ kPa} \\ T_2 = 70^\circ\text{C} \end{array} \right\} \nu_2 = 0.027413 \text{ m}^3/\text{kg}$$

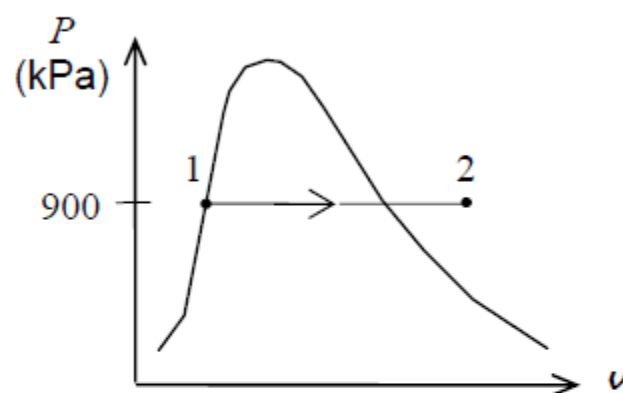
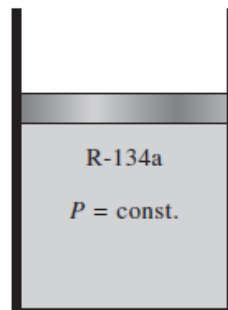
Analysis The boundary work is determined from its definition to be

$$m = \frac{V_1}{\nu_1} = \frac{0.2 \text{ m}^3}{0.0008580 \text{ m}^3/\text{kg}} = 233.1 \text{ kg}$$

and

$$\begin{aligned} W_{b,\text{out}} &= \int_1^2 P dV = P(V_2 - V_1) = mP(\nu_2 - \nu_1) \\ &= (233.1 \text{ kg})(900 \text{ kPa})(0.027413 - 0.0008580) \text{ m}^3/\text{kg} \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= \mathbf{5571 \text{ kJ}} \end{aligned}$$

Discussion The positive sign indicates that work is done by the system (work output).



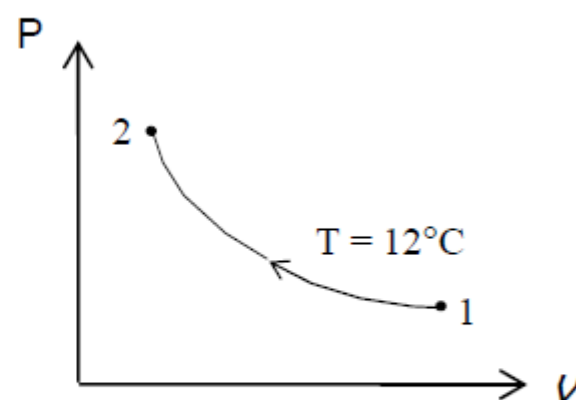
4-12 A mass of 2.4 kg of air at 150 kPa and 12°C is contained in a gas-tight, frictionless piston–cylinder device. The air is now compressed to a final pressure of 600 kPa. During the process, heat is transferred from the air such that the temperature inside the cylinder remains constant. Calculate the work input during this process. *Answer: 272 kJ*

Assumptions 1 The process is quasi-equilibrium. 2 Air is an ideal gas.

Properties The gas constant of air is $R = 0.287 \text{ kJ/kg}\cdot\text{K}$ (Table A-1).

Analysis The boundary work is determined from its definition to be

$$\begin{aligned} W_{b,\text{out}} &= \int_1^2 P dV = P_1 V_1 \ln \frac{V_2}{V_1} = mRT \ln \frac{P_1}{P_2} \\ &= (2.4 \text{ kg})(0.287 \text{ kJ/kg}\cdot\text{K})(285 \text{ K}) \ln \frac{150 \text{ kPa}}{600 \text{ kPa}} \\ &= -272 \text{ kJ} \end{aligned}$$



Discussion The negative sign indicates that work is done on the system (work input).

4-16

During some actual expansion and compression processes in piston–cylinder devices, the gases have been observed to satisfy the relationship $PV^n = C$, where n and C are constants. Calculate the work done when a gas expands from 150 kPa and 0.03 m^3 to a final volume of 0.2 m^3 for the case of $n = 1.3$.

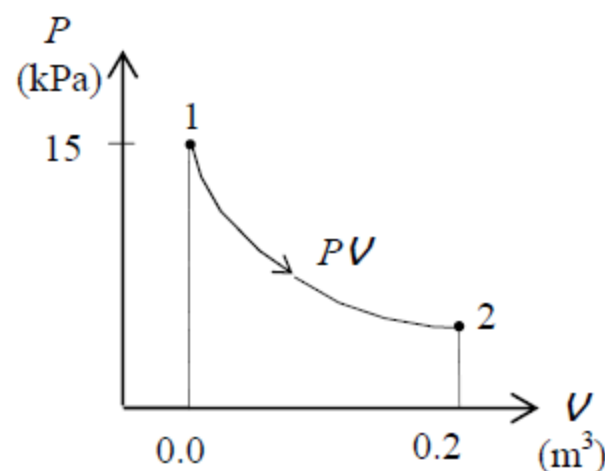
Assumptions The process is quasi-equilibrium.

Analysis The boundary work for this polytropic process can be determined directly from

$$P_2 = P_1 \left(\frac{V_1}{V_2} \right)^n = (150 \text{ kPa}) \left(\frac{0.03 \text{ m}^3}{0.2 \text{ m}^3} \right)^{1.3} = 12.74 \text{ kPa}$$

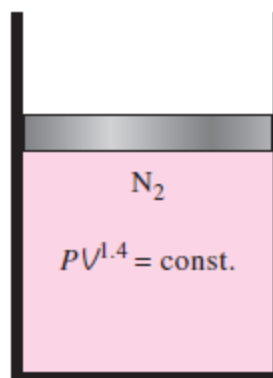
and,

$$\begin{aligned} W_{b,\text{out}} &= \int_1^2 P dV = \frac{P_2 V_2 - P_1 V_1}{1 - n} \\ &= \frac{(12.74 \times 0.2 - 150 \times 0.03) \text{ kPa} \cdot \text{m}^3}{1 - 1.3} \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= 6.51 \text{ kJ} \end{aligned}$$



Discussion The positive sign indicates that work is done by the system (work output).

4-18 A frictionless piston–cylinder device contains 2 kg of nitrogen at 100 kPa and 300 K. Nitrogen is now compressed slowly according to the relation $PV^{1.4} = \text{constant}$ until it reaches a final temperature of 360 K. Calculate the work input during this process. *Answer: 89 kJ*



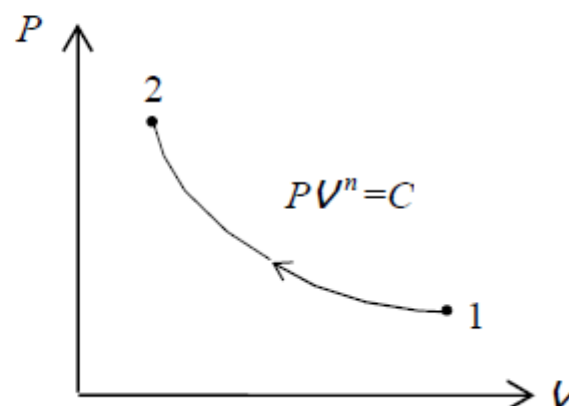
Assumptions 1 The process is quasi-equilibrium. 2 Nitrogen is an ideal gas.

Properties The gas constant for nitrogen is $R = 0.2968 \text{ kJ/kg}\cdot\text{K}$ (Table A-2a)

Analysis The boundary work for this polytropic process can be determined from

$$\begin{aligned} W_{b,\text{out}} &= \int_1^2 P dV = \frac{P_2 V_2 - P_1 V_1}{1 - n} = \frac{mR(T_2 - T_1)}{1 - n} \\ &= \frac{(2 \text{ kg})(0.2968 \text{ kJ/kg}\cdot\text{K})(360 - 300)\text{K}}{1 - 1.4} \\ &= -89.0 \text{ kJ} \end{aligned}$$

Discussion The negative sign indicates that work is done on the system (work input).



4-19

The equation of state of a gas is given as $\bar{v}(P + 10/\bar{v}^2) = R_u T$, where the units of $\bar{v}$ and P are m^3/kmol and kPa , respectively. Now 0.5 kmol of this gas is expanded in a quasi-equilibrium manner from 2 to 4 m^3 at a constant temperature of 300 K . Determine (a) the unit of the quantity 10 in the equation and (b) the work done during this isothermal expansion process.

4-21

Carbon dioxide contained in a piston–cylinder device is compressed from 0.3 to 0.1 m^3 . During the process, the pressure and volume are related by $P = aV^{-2}$, where $a = 8 \text{ kPa} \cdot \text{m}^6$. Calculate the work done on the carbon dioxide during this process. *Answer: 53.3 kJ*

4-26

A piston–cylinder device initially contains 0.25 kg of nitrogen gas at 130 kPa and 120°C . The nitrogen is now expanded isothermally to a pressure of 100 kPa . Determine the boundary work done during this process.

Answer: 7.65 kJ

4-27

A piston–cylinder device contains 0.15 kg of air initially at 2 MPa and 350°C . The air is first expanded isothermally to 500 kPa , then compressed polytropically with a polytropic exponent of 1.2 to the initial pressure, and finally compressed at the constant pressure to the initial state. Determine the boundary work for each process and the net work of the cycle.

4-34 A piston–cylinder device contains 5 kg of refrigerant-134a at 800 kPa and 70°C. The refrigerant is now cooled at constant pressure until it exists as a liquid at 15°C. Determine the amount of heat loss and show the process on a T - ν diagram with respect to saturation lines. *Answer: 1173 kJ*

Assumptions 1 The cylinder is stationary and thus the kinetic and potential energy changes are zero. 2 There are no work interactions involved other than the boundary work. 3 The thermal energy stored in the cylinder itself is negligible. 4 The compression or expansion process is quasi-equilibrium.

Analysis We take the contents of the cylinder as the system. This is a closed system since no mass enters or leaves. The energy balance for this stationary closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$-Q_{\text{out}} - W_{b,\text{out}} = \Delta U = m(u_2 - u_1) \quad (\text{since KE} = \text{PE} = 0)$$

$$-Q_{\text{out}} = m(h_2 - h_1)$$

since $\Delta U + W_b = \Delta H$ during a constant pressure quasi-equilibrium process. The properties of R-134a are

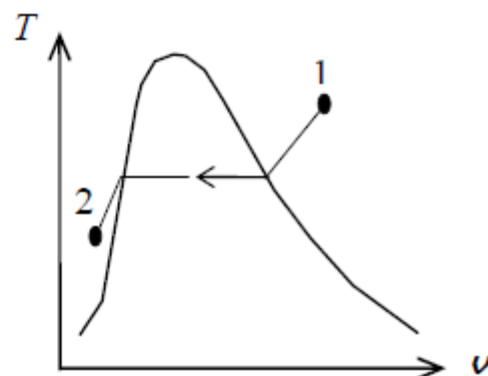
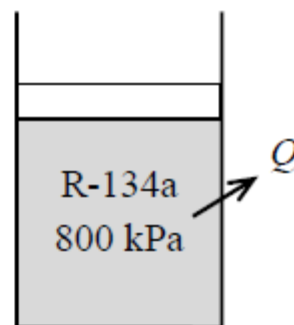
(Tables A-11 through A-13)

$$\left. \begin{array}{l} P_1 = 800 \text{ kPa} \\ T_1 = 70^\circ\text{C} \end{array} \right\} h_1 = 306.88 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_2 = 800 \text{ kPa} \\ T_2 = 15^\circ\text{C} \end{array} \right\} h_2 = h_{f@15^\circ\text{C}} = 72.34 \text{ kJ/kg}$$

Substituting,

$$Q_{\text{out}} = - (5 \text{ kg})(72.34 - 306.88) \text{ kJ/kg} = \mathbf{1173 \text{ kJ}}$$



4-36 An insulated piston–cylinder device contains 5 L of saturated liquid water at a constant pressure of 175 kPa. Water is stirred by a paddle wheel while a current of 8 A flows for 45 min through a resistor placed in the water. If one-half of the liquid is evaporated during this constant-pressure process and the paddle-wheel work amounts to 400 kJ, determine the voltage of the source. Also, show the process on a P - ν diagram with respect to saturation lines.

Answer: 224 V

Analysis We take the contents of the cylinder as the system. This is a closed system since no mass enters or leaves. The energy balance for this stationary closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{\text{system}}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc. energies}}}$$

$$W_{e,\text{in}} + W_{\text{pw},\text{in}} - W_{b,\text{out}} = \Delta U \quad (\text{since } Q = \text{KE} = \text{PE} = 0)$$

$$W_{e,\text{in}} + W_{\text{pw},\text{in}} = m(h_2 - h_1)$$

$$(VI\Delta t) + W_{\text{pw},\text{in}} = m(h_2 - h_1)$$

since $\Delta U + W_b = \Delta H$ during a constant pressure quasi-equilibrium process. The properties of water are (Tables A-4 through A-6)

$$\left. \begin{array}{l} P_1 = 175 \text{ kPa} \\ \text{sat. liquid} \end{array} \right\} \begin{array}{l} h_1 = h_f @ 175 \text{ kPa} = 487.01 \text{ kJ/kg} \\ \nu_1 = \nu_f @ 175 \text{ kPa} = 0.001057 \text{ m}^3/\text{kg} \end{array}$$

$$\left. \begin{array}{l} P_2 = 175 \text{ kPa} \\ x_2 = 0.5 \end{array} \right\} h_2 = h_f + x_2 h_{fg} = 487.01 + (0.5 \times 2213.1) = 1593.6 \text{ kJ/kg}$$

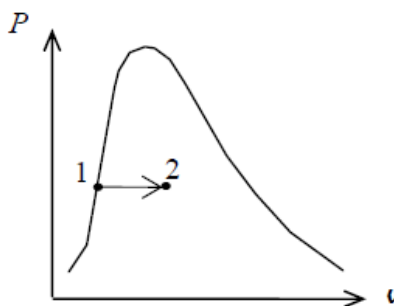
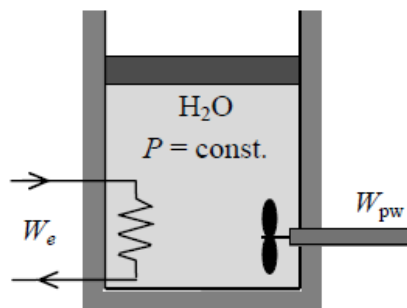
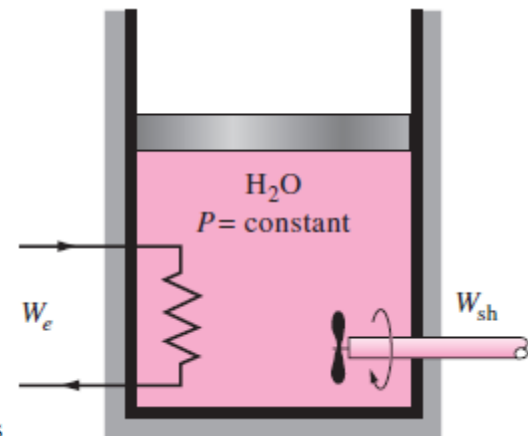
$$m = \frac{\nu_1}{\nu_1} = \frac{0.005 \text{ m}^3}{0.001057 \text{ m}^3/\text{kg}} = 4.731 \text{ kg}$$

Substituting,

$$VI\Delta t + (400 \text{ kJ}) = (4.731 \text{ kg})(1593.6 - 487.01) \text{ kJ/kg}$$

$$VI\Delta t = 4835 \text{ kJ}$$

$$V = \frac{4835 \text{ kJ}}{(8 \text{ A})(45 \times 60 \text{ s})} \left(\frac{1000 \text{ VA}}{1 \text{ kJ/s}} \right) = \mathbf{223.9 \text{ V}}$$



4-38



A piston–cylinder device initially contains steam at 200 kPa, 200°C, and 0.5 m³. At this state, a linear spring ($F \propto x$) is touching the piston but exerts no force on it. Heat is now slowly transferred to the steam, causing the pressure and the volume to rise to 500 kPa and 0.6 m³, respectively. Show the process on a P - v diagram with respect to saturation lines and determine (a) the final temperature, (b) the work done by the steam, and (c) the total heat transferred. **Answers: (a) 1132°C, (b) 35 kJ, (c) 808 kJ**

Analysis We take the contents of the cylinder as the system. This is a closed system since no mass enters or leaves. Noting that the spring is not part of the system (it is external), the energy balance for this stationary closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$Q_{\text{in}} - W_{b,\text{out}} = \Delta U = m(u_2 - u_1) \quad (\text{since KE} = \text{PE} = 0)$$

$$Q_{\text{in}} = m(u_2 - u_1) + W_{b,\text{out}}$$

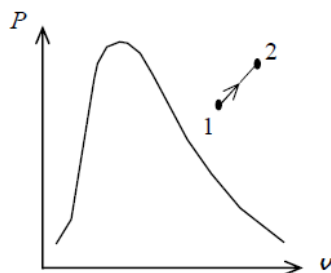
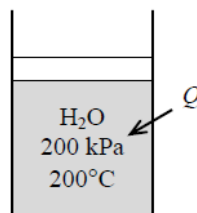
The properties of steam are (Tables A-4 through A-6)

$$\left. \begin{array}{l} P_1 = 200 \text{ kPa} \\ T_1 = 200^\circ\text{C} \end{array} \right\} \begin{array}{l} v_1 = 1.08049 \text{ m}^3/\text{kg} \\ u_1 = 2654.6 \text{ kJ/kg} \end{array}$$

$$m = \frac{V_1}{v_1} = \frac{0.5 \text{ m}^3}{1.08049 \text{ m}^3/\text{kg}} = 0.4628 \text{ kg}$$

$$v_2 = \frac{V_2}{m} = \frac{0.6 \text{ m}^3}{0.4628 \text{ kg}} = 1.2966 \text{ m}^3/\text{kg}$$

$$\left. \begin{array}{l} P_2 = 500 \text{ kPa} \\ v_2 = 1.2966 \text{ m}^3/\text{kg} \end{array} \right\} \begin{array}{l} T_2 = \mathbf{1132^\circ\text{C}} \\ u_2 = 4325.2 \text{ kJ/kg} \end{array}$$



(b) The pressure of the gas changes linearly with volume, and thus the process curve on a P - V diagram will be a straight line. The boundary work during this process is simply the area under the process curve, which is a trapezoidal. Thus,

$$W_b = \text{Area} = \frac{P_1 + P_2}{2} (V_2 - V_1) = \frac{(200 + 500) \text{ kPa}}{2} (0.6 - 0.5) \text{ m}^3 \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) = \mathbf{35 \text{ kJ}}$$

(c) From the energy balance we have

$$Q_{\text{in}} = (0.4628 \text{ kg})(4325.2 - 2654.6) \text{ kJ/kg} + 35 \text{ kJ} = \mathbf{808 \text{ kJ}}$$

4-41 Two tanks (Tank A and Tank B) are separated by a partition. Initially Tank A contains 2-kg steam at 1 MPa and 300°C while Tank B contains 3-kg saturated liquid–vapor mixture with a vapor mass fraction of 50 percent. Now the partition is removed and the two sides are allowed to mix until the mechanical and thermal equilibrium are established. If the pressure at the final state is 300 kPa, determine (a) the temperature and quality of the steam (if mixture) at the final state and (b) the amount of heat lost from the tanks.

Assumptions 1 The tank is stationary and thus the kinetic and potential energy changes are zero. 2 There are no work interactions.

Analysis (a) We take the contents of both tanks as the system. This is a closed system since no mass enters or leaves. Noting that the volume of the system is constant and thus there is no boundary work, the energy balance for this stationary closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$-Q_{\text{out}} = \Delta U_A + \Delta U_B = [m(u_2 - u_1)]_A + [m(u_2 - u_1)]_B \quad (\text{since } W = \text{KE} = \text{PE} = 0)$$

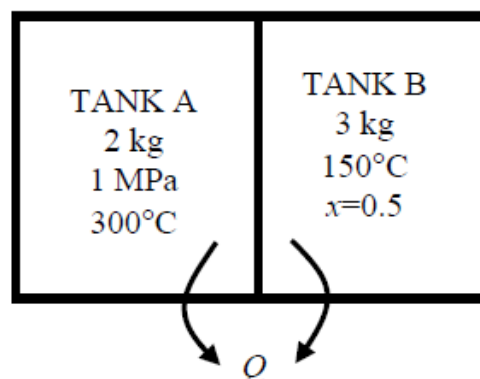
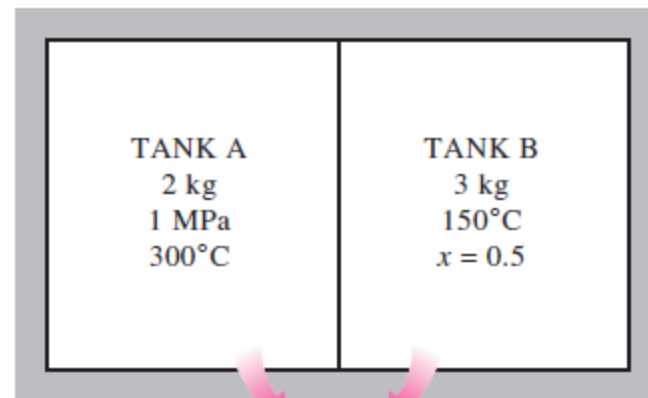
The properties of steam in both tanks at the initial state are (Tables A-4 through A-6)

$$\left. \begin{array}{l} P_{1,A} = 1000 \text{ kPa} \\ T_{1,A} = 300^\circ\text{C} \end{array} \right\} \begin{array}{l} v_{1,A} = 0.25799 \text{ m}^3/\text{kg} \\ u_{1,A} = 2793.7 \text{ kJ/kg} \end{array}$$

$$\left. \begin{array}{l} T_{1,B} = 150^\circ\text{C} \\ x_1 = 0.50 \end{array} \right\} \begin{array}{l} v_f = 0.001091, \quad v_g = 0.39248 \text{ m}^3/\text{kg} \\ u_f = 631.66, \quad u_{fg} = 1927.4 \text{ kJ/kg} \end{array}$$

$$v_{1,B} = v_f + x_1 v_{fg} = 0.001091 + [0.50 \times (0.39248 - 0.001091)] = 0.19679 \text{ m}^3/\text{kg}$$

$$u_{1,B} = u_f + x_1 u_{fg} = 631.66 + (0.50 \times 1927.4) = 1595.4 \text{ kJ/kg}$$



4-41 Continued

The total volume and total mass of the system are

$$V = V_A + V_B = m_A v_{1,A} + m_B v_{1,B} = (2 \text{ kg})(0.25799 \text{ m}^3/\text{kg}) + (3 \text{ kg})(0.19679 \text{ m}^3/\text{kg}) = 1.106 \text{ m}^3$$

$$m = m_A + m_B = 3 + 2 = 5 \text{ kg}$$

Now, the specific volume at the final state may be determined

$$v_2 = \frac{V}{m} = \frac{1.106 \text{ m}^3}{5 \text{ kg}} = 0.22127 \text{ m}^3/\text{kg}$$

which fixes the final state and we can determine other properties

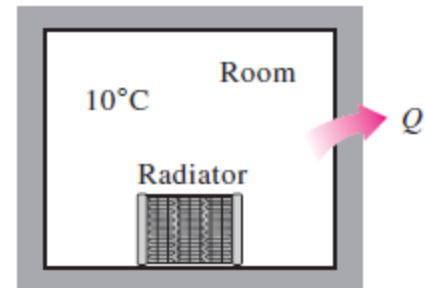
$$\begin{aligned} P_2 &= 300 \text{ kPa} \\ v_2 &= 0.22127 \text{ m}^3/\text{kg} \end{aligned} \left\{ \begin{aligned} T_2 &= T_{\text{sat @ } 300 \text{ kPa}} = \mathbf{133.5^\circ\text{C}} \\ x_2 &= \frac{v_2 - v_f}{v_g - v_f} = \frac{0.22127 - 0.001073}{0.60582 - 0.001073} = \mathbf{0.3641} \\ u_2 &= u_f + x_2 u_{fg} = 561.11 + (0.3641 \times 1982.1) = 1282.8 \text{ kJ/kg} \end{aligned} \right.$$

(b) Substituting,

$$\begin{aligned} -Q_{\text{out}} &= \Delta U_A + \Delta U_B = [m(u_2 - u_1)]_A + [m(u_2 - u_1)]_B \\ &= (2 \text{ kg})(1282.8 - 2793.7) \text{ kJ/kg} + (3 \text{ kg})(1282.8 - 1595.4) \text{ kJ/kg} = -3959 \text{ kJ} \end{aligned}$$

or $Q_{\text{out}} = \mathbf{3959 \text{ kJ}}$

4–42 A 30-L electrical radiator containing heating oil is placed in a 50-m³ room. Both the room and the oil in the radiator are initially at 10°C. The radiator with a rating of 1.8 kW is now turned on. At the same time, heat is lost from the room at an average rate of 0.35 kJ/s. After some time, the average temperature is measured to be 20°C for the air in the room, and 50°C for the oil in the radiator. Taking the density and the specific heat of the oil to be 950 kg/m³ and 2.2 kJ/kg · °C, respectively, determine how long the heater is kept on. Assume the room is well-sealed so that there are no air leaks.



Specific Heats, Δu , and Δh of Ideal Gases

4–53 Determine the internal energy change Δu of hydrogen, in kJ/kg, as it is heated from 200 to 800 K, using (a) the empirical specific heat equation as a function of temperature (Table A–2c), (b) the c_v value at the average temperature (Table A–2b), and (c) the c_v value at room temperature (Table A–2a).

4-57 A $4\text{-m} \times 5\text{-m} \times 6\text{-m}$ room is to be heated by a base-board resistance heater. It is desired that the resistance heater be able to raise the air temperature in the room from 7 to 23°C within 15 min. Assuming no heat losses from the room and an atmospheric pressure of 100 kPa, determine the required power of the resistance heater. Assume constant specific heats at room temperature. *Answer: 1.91 kW*

Analysis We take the air in the room to be the system. This is a closed system since no mass crosses the system boundary. The energy balance for this stationary constant-volume closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{\text{system}}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc. energies}}}$$

$$W_{e,\text{in}} = \Delta U \cong mc_{v,\text{avg}}(T_2 - T_1) \quad (\text{since } Q = \text{KE} = \text{PE} = 0)$$

or,

$$\dot{W}_{e,\text{in}} \Delta t = mc_{v,\text{avg}}(T_2 - T_1)$$

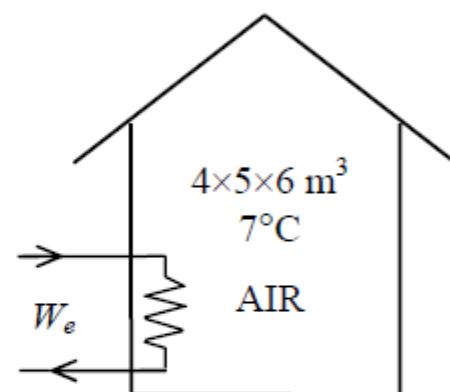
The mass of air is

$$V = 4 \times 5 \times 6 = 120 \text{ m}^3$$

$$m = \frac{P_1 V}{RT_1} = \frac{(100 \text{ kPa})(120 \text{ m}^3)}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(280 \text{ K})} = 149.3 \text{ kg}$$

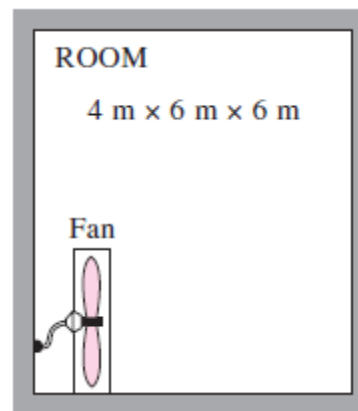
Substituting, the power rating of the heater becomes

$$\dot{W}_{e,\text{in}} = \frac{(149.3 \text{ kg})(0.718 \text{ kJ/kg} \cdot ^\circ\text{C})(23 - 7)^\circ\text{C}}{15 \times 60 \text{ s}} = \mathbf{1.91 \text{ kW}}$$



4–59 A student living in a $4\text{-m} \times 6\text{-m} \times 6\text{-m}$ dormitory room turns on her 150-W fan before she leaves the room on a summer day, hoping that the room will be cooler when she comes back in the evening. Assuming all the doors and windows are tightly closed and disregarding any heat transfer through the walls and the windows, determine the temperature in the room when she comes back 10 h later. Use specific heat values at room temperature, and assume the room to be at 100 kPa and 15°C in the morning when she leaves.

Answer: 58.2°C



4-62 A piston–cylinder device whose piston is resting on top of a set of stops initially contains 0.5 kg of helium gas at 100 kPa and 25°C. The mass of the piston is such that 500 kPa of pressure is required to raise it. How much heat must be transferred to the helium before the piston starts rising?

Answer: 1857 kJ

Analysis We take the helium gas in the cylinder as the system. This is a closed system since no mass crosses the boundary of the system. The energy balance for this constant volume closed system can be expressed as

$$\underbrace{E_{in} - E_{out}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$Q_{in} = \Delta U = m(u_2 - u_1)$$

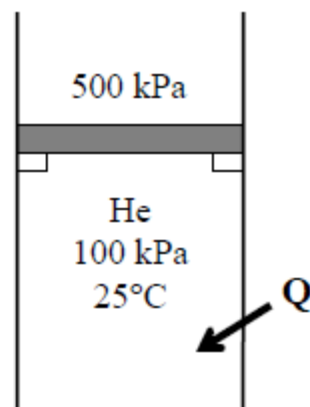
$$Q_{in} = m(u_2 - u_1) = mc_v(T_2 - T_1)$$

The final temperature of helium can be determined from the ideal gas relation to be

$$\frac{P_1 V}{T_1} = \frac{P_2 V}{T_2} \longrightarrow T_2 = \frac{P_2}{P_1} T_1 = \frac{500 \text{ kPa}}{100 \text{ kPa}} (298 \text{ K}) = 1490 \text{ K}$$

Substituting into the energy balance relation gives

$$Q_{in} = (0.5 \text{ kg})(3.1156 \text{ kJ/kg}\cdot\text{K})(1490 - 298)\text{K} = \mathbf{1857 \text{ kJ}}$$



4-65 A mass of 15 kg of air in a piston–cylinder device is heated from 25 to 77°C by passing current through a resistance heater inside the cylinder. The pressure inside the cylinder is held constant at 300 kPa during the process, and a heat loss of 60 kJ occurs. Determine the electric energy supplied, in kWh.

Answer: 0.235 kWh

Properties The initial and final enthalpies of air are (Table A-17)

$$h_1 = h_{@298\text{ K}} = 298.18 \text{ kJ/kg}$$

$$h_2 = h_{@350\text{ K}} = 350.49 \text{ kJ/kg}$$

Analysis We take the contents of the cylinder as the system. This is a closed system since no mass enters or leaves. The energy balance for this closed system can be expressed as

$$\underbrace{E_{in} - E_{out}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$W_{e,in} - Q_{out} - W_{b,out} = \Delta U \longrightarrow W_{e,in} = m(h_2 - h_1) + Q_{out}$$

since $\Delta U + W_b = \Delta H$ during a constant pressure quasi-equilibrium process. Substituting,

$$W_{e,in} = (15 \text{ kg})(350.49 - 298.18) \text{ kJ/kg} + (60 \text{ kJ}) = 845 \text{ kJ}$$

or,

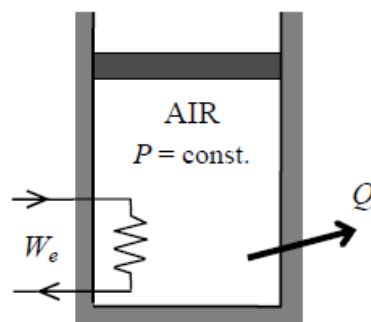
$$W_{e,in} = (845 \text{ kJ}) \left(\frac{1 \text{ kWh}}{3600 \text{ kJ}} \right) = 0.235 \text{ kWh}$$

Alternative solution The specific heat of air at the average temperature of $T_{\text{avg}} = (25 + 77)/2 = 51^\circ\text{C} = 324 \text{ K}$ is, from Table A-2b, $c_{p,\text{avg}} = 1.0065 \text{ kJ/kg}\cdot^\circ\text{C}$. Substituting,

$$W_{e,in} = mc_p(T_2 - T_1) + Q_{out} = (15 \text{ kg})(1.0065 \text{ kJ/kg}\cdot^\circ\text{C})(77 - 25)^\circ\text{C} + 60 \text{ kJ} = 845 \text{ kJ}$$

or,

$$W_{e,in} = (845 \text{ kJ}) \left(\frac{1 \text{ kWh}}{3600 \text{ kJ}} \right) = 0.235 \text{ kWh}$$



4-66 An insulated piston–cylinder device initially contains 0.3 m^3 of carbon dioxide at 200 kPa and 27°C . An electric switch is turned on, and a 110-V source supplies current to a resistance heater inside the cylinder for a period of 10 min . The pressure is held constant during the process, while the volume is doubled. Determine the current that passes through the resistance heater.

4-67 A piston–cylinder device contains 0.8 kg of nitrogen initially at 100 kPa and 27°C. The nitrogen is now compressed slowly in a polytropic process during which $PV^{1.3} = \text{constant}$ until the volume is reduced by one-half. Determine the work done and the heat transfer for this process.

Analysis We take the contents of the cylinder as the system. This is a closed system since no mass crosses the system boundary. The energy balance for this closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{\text{system}}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc. energies}}}$$

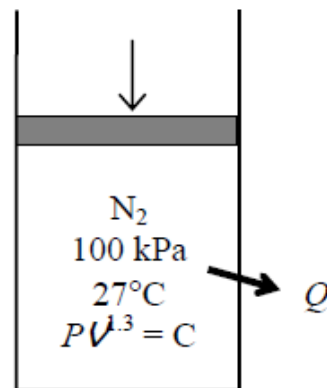
$$W_{\text{b,in}} - Q_{\text{out}} = \Delta U = m(u_2 - u_1)$$

$$W_{\text{b,in}} - Q_{\text{out}} = mc_v(T_2 - T_1)$$

The final pressure and temperature of nitrogen are

$$P_2 V_2^{1.3} = P_1 V_1^{1.3} \longrightarrow P_2 = \left(\frac{V_1}{V_2} \right)^{1.3} P_1 = 2^{1.3} (100 \text{ kPa}) = 246.2 \text{ kPa}$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \longrightarrow T_2 = \frac{P_2 V_2}{P_1 V_1} T_1 = \frac{246.2 \text{ kPa}}{100 \text{ kPa}} \times 0.5 \times (300 \text{ K}) = 369.3 \text{ K}$$



Then the boundary work for this polytropic process can be determined from

$$\begin{aligned} W_{\text{b,in}} &= - \int_1^2 P dV = - \frac{P_2 V_2 - P_1 V_1}{1 - n} = - \frac{mR(T_2 - T_1)}{1 - n} \\ &= - \frac{(0.8 \text{ kg})(0.2968 \text{ kJ/kg} \cdot \text{K})(369.3 - 300) \text{ K}}{1 - 1.3} = \mathbf{54.8 \text{ kJ}} \end{aligned}$$

Substituting into the energy balance gives

$$\begin{aligned} Q_{\text{out}} &= W_{\text{b,in}} - mc_v(T_2 - T_1) \\ &= 54.8 \text{ kJ} - (0.8 \text{ kg})(0.744 \text{ kJ/kg} \cdot \text{K})(369.3 - 300) \text{ K} \\ &= \mathbf{13.6 \text{ kJ}} \end{aligned}$$

4–69 A room is heated by a baseboard resistance heater. When the heat losses from the room on a winter day amount to 6500 kJ/h, the air temperature in the room remains constant even though the heater operates continuously. Determine the power rating of the heater, in kW.

4-73



A piston–cylinder device, with a set of stops on the top, initially contains 3 kg of air at 200 kPa and 27°C. Heat is now transferred to the air, and the piston rises until it hits the stops, at which point the volume is twice the initial volume. More heat is transferred until the pressure inside the cylinder also doubles. Determine the work done and the amount of heat transfer for this process. Also, show the process on a P - v diagram.

Analysis We take the air in the cylinder to be the system. This is a closed system since no mass crosses the boundary of the system. The energy balance for this closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$Q_{\text{in}} - W_{\text{b,out}} = \Delta U = m(u_3 - u_1)$$

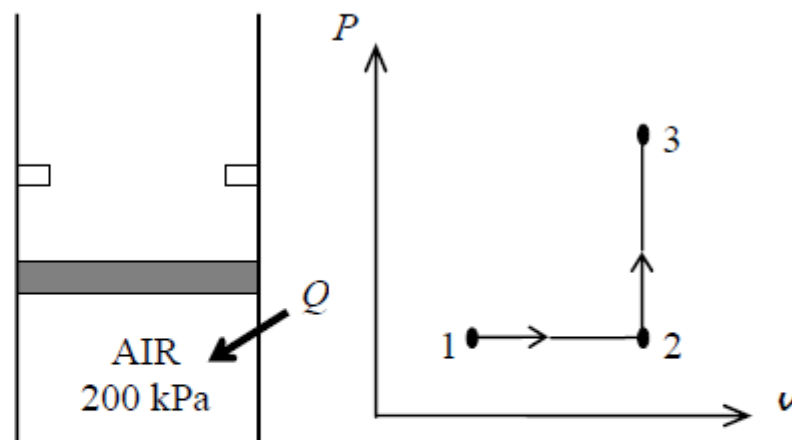
$$Q_{\text{in}} = m(u_3 - u_1) + W_{\text{b,out}}$$

The initial and the final volumes and the final temperature of air are determined from

$$V_1 = \frac{mRT_1}{P_1} = \frac{(3 \text{ kg})(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(300 \text{ K})}{200 \text{ kPa}} = 1.29 \text{ m}^3$$

$$V_3 = 2V_1 = 2 \times 1.29 = 2.58 \text{ m}^3$$

$$\frac{P_1 V_1}{T_1} = \frac{P_3 V_3}{T_3} \longrightarrow T_3 = \frac{P_3 V_3}{P_1 V_1} T_1 = \frac{400 \text{ kPa}}{200 \text{ kPa}} \times 2 \times (300 \text{ K}) = 1200 \text{ K}$$



4-73 Continued

No work is done during process 2-3 since $V_2 = V_3$. The pressure remains constant during process 1-2 and the work done during this process is

$$\begin{aligned} W_b &= \int_1^2 P dV = P_2 (V_3 - V_2) \\ &= (200 \text{ kPa})(2.58 - 1.29) \text{ m}^3 \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) = \mathbf{258 \text{ kJ}} \end{aligned}$$

The initial and final internal energies of air are (Table A-17)

$$u_1 = u_{@300 \text{ K}} = 214.07 \text{ kJ/kg}$$

$$u_2 = u_{@1200 \text{ K}} = 933.33 \text{ kJ/kg}$$

Substituting,

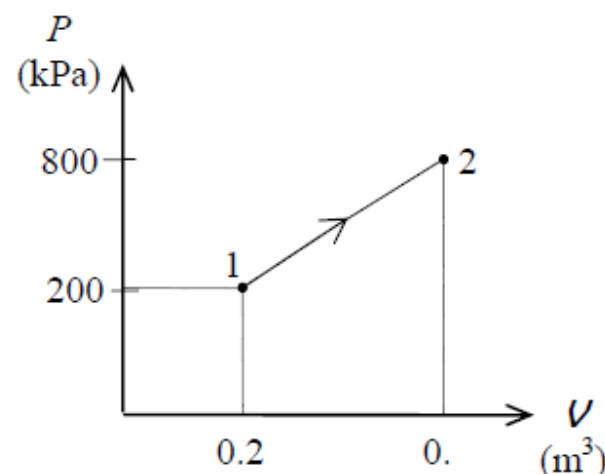
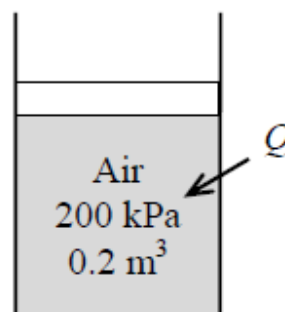
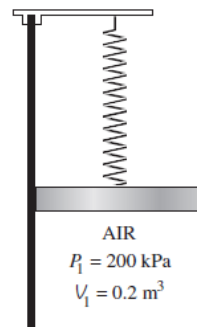
$$Q_{\text{in}} = (3 \text{ kg})(933.33 - 214.07) \text{ kJ/kg} + 258 \text{ kJ} = \mathbf{2416 \text{ kJ}}$$

Alternative solution The specific heat of air at the average temperature of $T_{\text{avg}} = (300 + 1200)/2 = 750 \text{ K}$ is, from Table A-2b, $c_{v,\text{avg}} = 0.800 \text{ kJ/kg}\cdot\text{K}$. Substituting

$$\begin{aligned} Q_{\text{in}} &= m(u_3 - u_1) + W_{b,\text{out}} \cong mc_v(T_3 - T_1) + W_{b,\text{out}} \\ &= (3 \text{ kg})(0.800 \text{ kJ/kg}\cdot\text{K})(1200 - 300) \text{ K} + 258 \text{ kJ} = \mathbf{2418 \text{ kJ}} \end{aligned}$$

4-114 A frictionless piston–cylinder device initially contains air at 200 kPa and 0.2 m³. At this state, a linear spring ($F \propto x$) is touching the piston but exerts no force on it. The air is now heated to a final state of 0.5 m³ and 800 kPa. Determine (a) the total work done by the air and (b) the work done against the spring. Also, show the process on a P - v diagram.

Answers: (a) 150 kJ, (b) 90 kJ



Analysis (a) The pressure of the gas changes linearly with volume during this process, and thus the process curve on a P - V diagram will be a straight line. Then the boundary work during this process is simply the area under the process curve, which is a trapezoidal. Thus,

$$\begin{aligned} W_{b,\text{out}} &= \text{Area} = \frac{P_1 + P_2}{2} (v_2 - v_1) \\ &= \frac{(200 + 800) \text{ kPa}}{2} (0.5 - 0.2) \text{ m}^3 \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= 150 \text{ kJ} \end{aligned}$$

(b) If there were no spring, we would have a constant pressure process at $P = 200$ kPa. The work done during this process is

$$\begin{aligned} W_{b,\text{out,no spring}} &= \int_1^2 P dv = P(v_2 - v_1) \\ &= (200 \text{ kPa}) (0.5 - 0.2) \text{ m}^3 / \text{kg} \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= 60 \text{ kJ} \end{aligned}$$

Thus,

$$W_{\text{spring}} = W_b - W_{b,\text{no spring}} = 150 - 60 = 90 \text{ kJ}$$

4-115 A mass of 5 kg of saturated liquid–vapor mixture of water is contained in a piston–cylinder device at 125 kPa. Initially, 2 kg of the water is in the liquid phase and the rest is in the vapor phase. Heat is now transferred to the water, and the piston, which is resting on a set of stops, starts moving when the pressure inside reaches 300 kPa. Heat transfer continues until the total volume increases by 20 percent. Determine (a) the initial and final temperatures, (b) the mass of liquid water when the piston first starts moving, and (c) the work done during this process. Also, show the process on a P - v diagram.

Assumptions The process is quasi-equilibrium.

Analysis (a) Initially the system is a saturated mixture at 125 kPa pressure, and thus the initial temperature is

$$T_1 = T_{\text{sat}@125 \text{ kPa}} = 106.0^\circ\text{C}$$

The total initial volume is

$$V_1 = m_f v_f + m_g v_g = 2 \times 0.001048 + 3 \times 1.3750 = 4.127 \text{ m}^3$$

Then the total and specific volumes at the final state are

$$V_3 = 1.2V_1 = 1.2 \times 4.127 = 4.953 \text{ m}^3$$

$$v_3 = \frac{V_3}{m} = \frac{4.953 \text{ m}^3}{5 \text{ kg}} = 0.9905 \text{ m}^3/\text{kg}$$

Thus,

$$\left. \begin{array}{l} P_3 = 300 \text{ kPa} \\ v_3 = 0.9905 \text{ m}^3/\text{kg} \end{array} \right\} T_3 = 373.6^\circ\text{C}$$

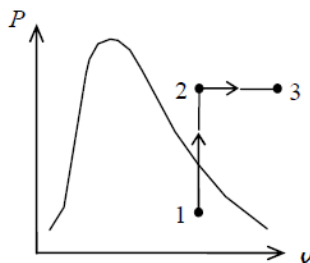
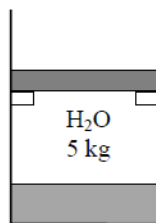
(b) When the piston first starts moving, $P_2 = 300 \text{ kPa}$ and $V_2 = V_1 = 4.127 \text{ m}^3$. The specific volume at this state is

$$v_2 = \frac{V_2}{m} = \frac{4.127 \text{ m}^3}{5 \text{ kg}} = 0.8254 \text{ m}^3/\text{kg}$$

which is greater than $v_g = 0.60582 \text{ m}^3/\text{kg}$ at 300 kPa. Thus **no liquid** is left in the cylinder when the piston starts moving.

(c) No work is done during process 1-2 since $V_1 = V_2$. The pressure remains constant during process 2-3 and the work done during this process is

$$W_b = \int_2^3 P dV = P_2 (V_3 - V_2) = (300 \text{ kPa})(4.953 - 4.127) \text{ m}^3 \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) = 247.6 \text{ kJ}$$



4-118 A mass of 12 kg of saturated refrigerant-134a vapor is contained in a piston–cylinder device at 240 kPa. Now 300 kJ of heat is transferred to the refrigerant at constant pressure while a 110-V source supplies current to a resistor within the cylinder for 6 min. Determine the current supplied if the final temperature is 70°C. Also, show the process on a T - v diagram with respect to the saturation lines. *Answer: 12.8 A*

4-119 A mass of 0.2 kg of saturated refrigerant-134a is contained in a piston–cylinder device at 200 kPa. Initially, 75 percent of the mass is in the liquid phase. Now heat is transferred to the refrigerant at constant pressure until the cylinder contains vapor only. Show the process on a P - ν diagram with respect to saturation lines. Determine (a) the volume occupied by the refrigerant initially, (b) the work done, and (c) the total heat transfer.

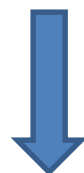
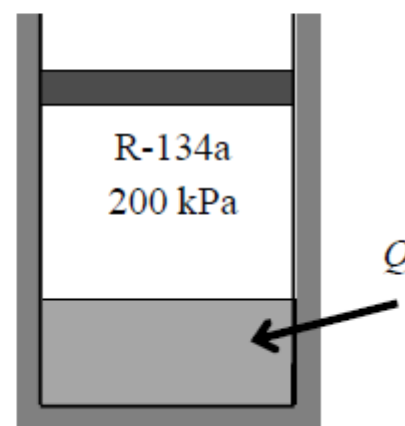
Analysis (a) Using property data from R-134a tables (Tables A-11 through A-13), the initial volume of the refrigerant is determined to be

$$\left. \begin{array}{l} P_1 = 200 \text{ kPa} \\ x_1 = 0.25 \end{array} \right\} \begin{array}{l} \nu_f = 0.0007533, \quad \nu_g = 0.099867 \text{ m}^3/\text{kg} \\ u_f = 38.28, \quad u_{fg} = 186.21 \text{ kJ/kg} \end{array}$$

$$\begin{aligned} \nu_1 &= \nu_f + x_1 \nu_{fg} \\ &= 0.0007533 + 0.25 \times (0.099867 - 0.0007533) = 0.02553 \text{ m}^3/\text{kg} \end{aligned}$$

$$u_1 = u_f + x_1 u_{fg} = 38.28 + 0.25 \times 186.21 = 84.83 \text{ kJ/kg}$$

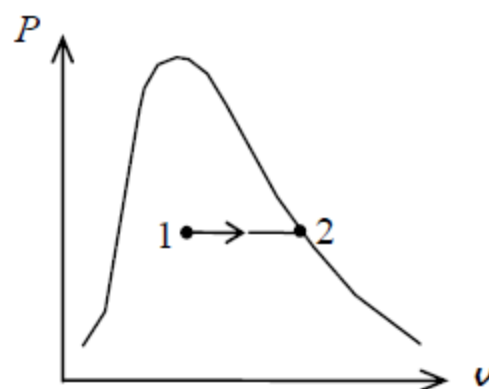
$$V_1 = m \nu_1 = (0.2 \text{ kg})(0.02553 \text{ m}^3/\text{kg}) = \mathbf{0.005106 \text{ m}^3}$$



(b) The work done during this constant pressure process is

$$\left. \begin{array}{l} P_2 = 200 \text{ kPa} \\ \text{sat. vapor} \end{array} \right\} \begin{array}{l} v_2 = v_{g@200 \text{ kPa}} = 0.09987 \text{ m}^3/\text{kg} \\ u_2 = u_{g@200 \text{ kPa}} = 224.48 \text{ kJ/kg} \end{array}$$

$$\begin{aligned} W_{b,\text{out}} &= \int_1^2 P dV = P(v_2 - v_1) = mP(v_2 - v_1) \\ &= (0.2 \text{ kg})(200 \text{ kPa})(0.09987 - 0.02553) \text{ m}^3/\text{kg} \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= \mathbf{2.97 \text{ kJ}} \end{aligned}$$



(c) We take the contents of the cylinder as the system. This is a closed system since no mass enters or leaves. The energy balance for this stationary closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

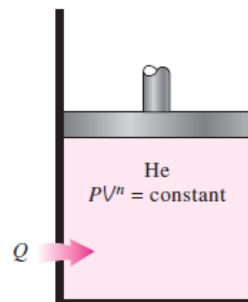
$$Q_{\text{in}} - W_{b,\text{out}} = \Delta U$$

$$Q_{\text{in}} = m(u_2 - u_1) + W_{b,\text{out}}$$

Substituting,

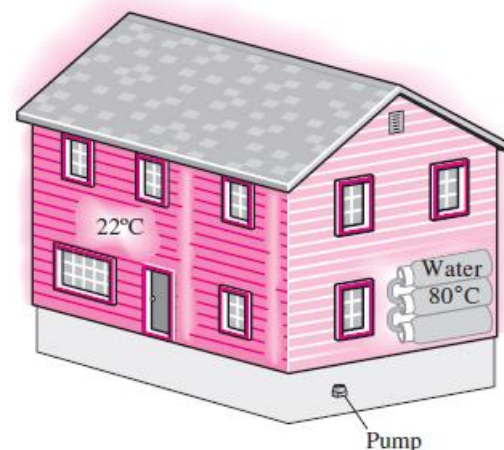
$$Q_{\text{in}} = (0.2 \text{ kg})(224.48 - 84.83) \text{ kJ/kg} + 2.97 = \mathbf{30.9 \text{ kJ}}$$

4-120 A piston–cylinder device contains helium gas initially at 150 kPa, 20°C, and 0.5 m³. The helium is now compressed in a polytropic process ($PV^n = \text{constant}$) to 400 kPa and 140°C. Determine the heat loss or gain during this process. *Answer: 11.2 kJ loss*



4-121 A frictionless piston–cylinder device and a rigid tank initially contain 12 kg of an ideal gas each at the same temperature, pressure, and volume. It is desired to raise the temperatures of both systems by 15°C. Determine the amount of extra heat that must be supplied to the gas in the cylinder which is maintained at constant pressure to achieve this result. Assume the molar mass of the gas is 25.

4-122 A passive solar house that is losing heat to the outdoors at an average rate of 50,000 kJ/h is maintained at 22°C at all times during a winter night for 10 h. The house is to be heated by 50 glass containers each containing 20 L of water that is heated to 80°C during the day by absorbing solar energy. A thermostat-controlled 15-kW back-up electric resistance heater turns on whenever necessary to keep the house at 22°C. (a) How long did the electric heating system run that night? (b) How long would the electric heater run that night if the house incorporated no solar heating? *Answers: (a) 4.77 h, (b) 9.26 h*



4-123 An 1800-W electric resistance heating element is immersed in 40 kg of water initially at 20°C. Determine how long it will take for this heater to raise the water temperature to 80°C.

4-124 One ton (1000 kg) of liquid water at 80°C is brought into a well-insulated and well-sealed 4-m × 5-m × 6-m room initially at 22°C and 100 kPa. Assuming constant specific heats for both air and water at room temperature, determine the final equilibrium temperature in the room. *Answer: 78.6°C*

4-125 A 4-m $\times$ 5-m $\times$ 6-m room is to be heated by one ton (1000 kg) of liquid water contained in a tank that is placed in the room. The room is losing heat to the outside at an average rate of 8000 kJ/h. The room is initially at 20°C and 100 kPa and is maintained at an average temperature of 20°C at all times. If the hot water is to meet the heating requirements of this room for a 24-h period, determine the minimum temperature of the water when it is first brought into the room. Assume constant specific heats for both air and water at room temperature.

Analysis Heat loss from the room during a 24-h period is

$$Q_{\text{loss}} = (8000 \text{ kJ/h})(24 \text{ h}) = 192,000 \text{ kJ}$$

Taking the contents of the room, including the water, as our system, the energy balance can be written as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$-Q_{\text{out}} = \Delta U = (\Delta U)_{\text{water}} + (\Delta U)_{\text{air}} \quad \text{?}$$

or

$$-Q_{\text{out}} = [mc(T_2 - T_1)]_{\text{water}}$$

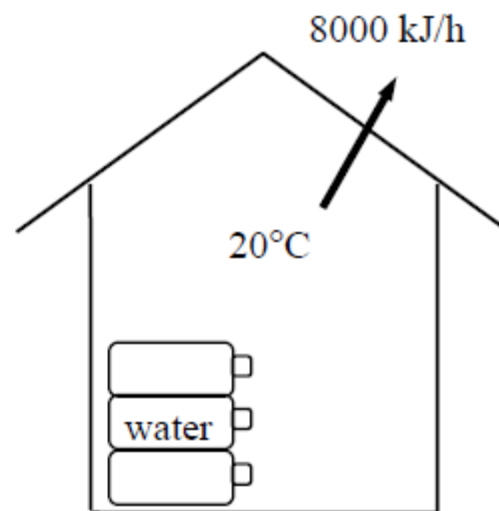
Substituting,

$$-192,000 \text{ kJ} = (1000 \text{ kg})(4.18 \text{ kJ/kg}\cdot^\circ\text{C})(20 - T_1)$$

It gives

$$T_1 = 65.9^\circ\text{C}$$

where T_1 is the temperature of the water when it is first brought into the room.



4-126 The energy content of a certain food is to be determined in a bomb calorimeter that contains 3 kg of water by burning a 2-g sample of it in the presence of 100 g of air in the reaction chamber. If the water temperature rises by 3.2°C when equilibrium is established, determine the energy content of the food, in kJ/kg, by neglecting the thermal energy stored in the reaction chamber and the energy supplied by the mixer. What is a rough estimate of the error involved in neglecting the thermal energy stored in the reaction chamber?

Answer: 20,060 kJ/kg

4-127 A 68-kg man whose average body temperature is 39°C drinks 1 L of cold water at 3°C in an effort to cool down. Taking the average specific heat of the human body to be $3.6 \text{ kJ/kg} \cdot ^{\circ}\text{C}$, determine the drop in the average body temperature of this person under the influence of this cold water.

4-128 A 0.2-L glass of water at 20°C is to be cooled with ice to 5°C. Determine how much ice needs to be added to the water, in grams, if the ice is at (a) 0°C and (b) -8°C. Also determine how much water would be needed if the cooling is to be done with cold water at 0°C. The melting temperature and the heat of fusion of ice at atmospheric pressure are 0°C and 333.7 kJ/kg, respectively, and the density of water is 1 kg/L.

Analysis (a) The mass of the water is

$$m_w = \rho V = (1 \text{ kg/L})(0.2 \text{ L}) = 0.2 \text{ kg}$$

We take the ice and the water as our system, and disregard any heat and mass transfer. This is a reasonable assumption since the time period of the process is very short. Then the energy balance can be written as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$0 = \Delta U$$

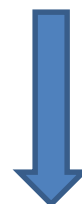
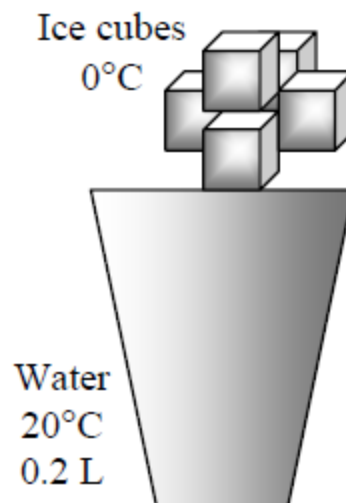
$$0 = \Delta U_{\text{ice}} + \Delta U_{\text{water}}$$

$$[mc(0^\circ\text{C} - T_1)_{\text{solid}} + mh_{\text{if}} + mc(T_2 - 0^\circ\text{C})_{\text{liquid}}]_{\text{ice}} + [mc(T_2 - T_1)]_{\text{water}} = 0$$

Noting that $T_{1,\text{ice}} = 0^\circ\text{C}$ and $T_2 = 5^\circ\text{C}$ and substituting gives

$$m[0 + 333.7 \text{ kJ/kg} + (4.18 \text{ kJ/kg}\cdot^\circ\text{C})(5 - 0)^\circ\text{C}] + (0.2 \text{ kg})(4.18 \text{ kJ/kg}\cdot^\circ\text{C})(5 - 20)^\circ\text{C} = 0$$

$$m = 0.0364 \text{ kg} = \mathbf{36.4 \text{ g}}$$



(b) When $T_{1,\text{ice}} = -8^\circ\text{C}$ instead of 0°C , substituting gives

$$m[(2.11 \text{ kJ/kg}\cdot^\circ\text{C})[0-(-8)]^\circ\text{C} + 333.7 \text{ kJ/kg} + (4.18 \text{ kJ/kg}\cdot^\circ\text{C})(5-0)^\circ\text{C}] \\ + (0.2 \text{ kg})(4.18 \text{ kJ/kg}\cdot^\circ\text{C})(5-20)^\circ\text{C} = 0$$

$$m = 0.0347 \text{ kg} = \mathbf{34.7 \text{ g}}$$

Cooling with cold water can be handled the same way. All we need to do is replace the terms for ice by a term for cold water at 0°C :

$$(\Delta U)_{\text{coldwater}} + (\Delta U)_{\text{water}} = 0$$

$$[mc(T_2 - T_1)]_{\text{coldwater}} + [mc(T_2 - T_1)]_{\text{water}} = 0$$

Substituting,

$$[m_{\text{cold water}} (4.18 \text{ kJ/kg}\cdot^\circ\text{C})(5 - 0)^\circ\text{C}] + (0.2 \text{ kg})(4.18 \text{ kJ/kg}\cdot^\circ\text{C})(5-20)^\circ\text{C} = 0$$

It gives

$$m = 0.6 \text{ kg} = \mathbf{600 \text{ g}}$$

Discussion Note that this is 17 times the amount of ice needed, and it explains why we use ice instead of water to cool drinks. Also, the temperature of ice does not seem to make a significant difference.

4-136 A rigid tank containing 0.4 m^3 of air at 400 kPa and 30°C is connected by a valve to a piston–cylinder device with zero clearance. The mass of the piston is such that a pressure of 200 kPa is required to raise the piston. The valve is now opened slightly, and air is allowed to flow into the cylinder until the pressure in the tank drops to 200 kPa . During this process, heat is exchanged with the surroundings such that the entire air remains at 30°C at all times. Determine the heat transfer for this process.

Analysis We take the entire air in the tank and the cylinder to be the system. This is a closed system since no mass crosses the boundary of the system. The energy balance for this closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$Q_{\text{in}} - W_{\text{b,out}} = \Delta U = m(u_2 - u_1) = 0$$

$$Q_{\text{in}} = W_{\text{b,out}}$$

since $u = u(T)$ for ideal gases, and thus $u_2 = u_1$ when $T_1 = T_2$. The initial volume of air is

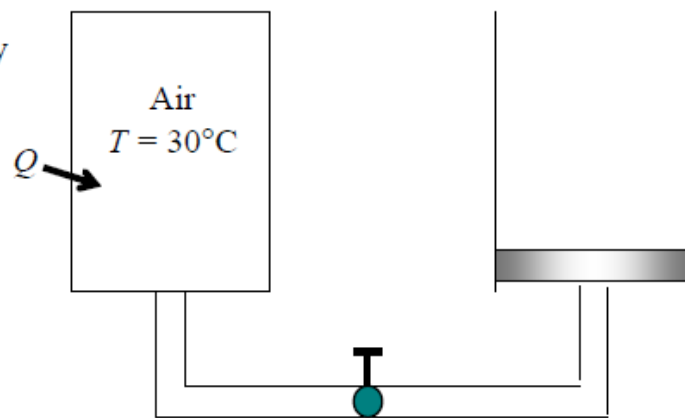
$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \longrightarrow V_2 = \frac{P_1}{P_2} \frac{T_2}{T_1} V_1 = \frac{400 \text{ kPa}}{200 \text{ kPa}} \times 1 \times (0.4 \text{ m}^3) = 0.80 \text{ m}^3$$

The pressure at the piston face always remains constant at 200 kPa . Thus the boundary work done during this process is

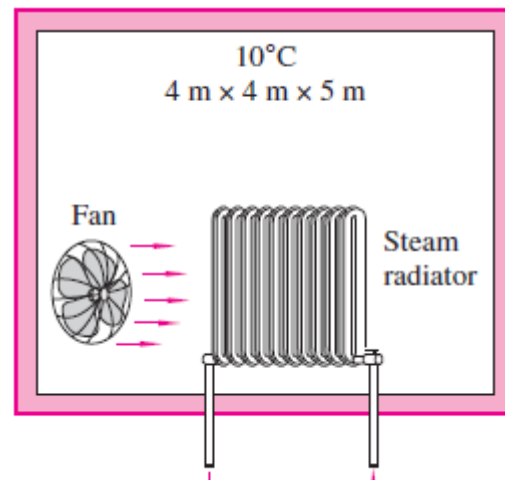
$$W_{\text{b,out}} = \int_1^2 P dV = P_2 (V_2 - V_1) = (200 \text{ kPa})(0.8 - 0.4) \text{ m}^3 \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) = 80 \text{ kJ}$$

Therefore, the heat transfer is determined from the energy balance to be

$$W_{\text{b,out}} = Q_{\text{in}} = \mathbf{80 \text{ kJ}}$$



4-137 A well-insulated 4-m × 4-m × 5-m room initially at 10°C is heated by the radiator of a steam heating system. The radiator has a volume of 15 L and is filled with superheated vapor at 200 kPa and 200°C. At this moment both the inlet and the exit valves to the radiator are closed. A 120-W fan is used to distribute the air in the room. The pressure of the steam is observed to drop to 100 kPa after 30 min as a result of heat transfer to the room. Assuming constant specific heats for air at room temperature, determine the average temperature of air in 30 min. Assume the air pressure in the room remains constant at 100 kPa.



Analysis We first take the radiator as the system. This is a closed system since no mass enters or leaves. The energy balance for this closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$-Q_{\text{out}} = \Delta U = m(u_2 - u_1) \quad (\text{since } W = \text{KE} = \text{PE} = 0)$$

$$Q_{\text{out}} = m(u_1 - u_2)$$

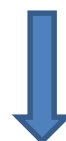
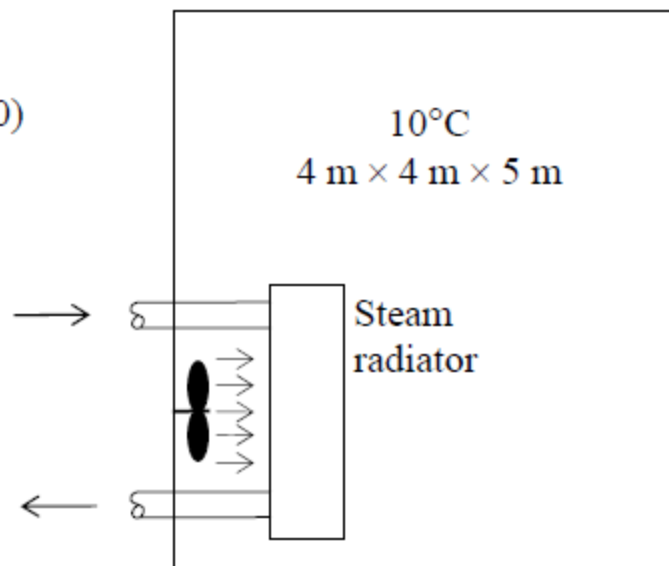
Using data from the steam tables (Tables A-4 through A-6), some properties are determined to be

$$P_1 = 200 \text{ kPa} \left\{ \begin{array}{l} \nu_1 = 1.08049 \text{ m}^3/\text{kg} \\ u_1 = 2654.6 \text{ kJ/kg} \end{array} \right.$$

$$T_1 = 200^\circ\text{C} \left\{ \begin{array}{l} \nu_1 = 1.08049 \text{ m}^3/\text{kg} \\ u_1 = 2654.6 \text{ kJ/kg} \end{array} \right.$$

$$P_2 = 100 \text{ kPa} \left\{ \begin{array}{l} \nu_f = 0.001043, \quad \nu_g = 1.6941 \text{ m}^3/\text{kg} \\ u_f = 417.40, \quad u_{fg} = 2088.2 \text{ kJ/kg} \end{array} \right. \quad (\nu_2 = \nu_1)$$

$$x_2 = \frac{\nu_2 - \nu_f}{\nu_{fg}} = \frac{1.08049 - 0.001043}{1.6941 - 0.001043} = 0.6376$$



$$u_2 = u_f + x_2 u_{fg} = 417.40 + 0.6376 \times 2088.2 = 1748.7 \text{ kJ/kg}$$

$$m = \frac{V_1}{v_1} = \frac{0.015 \text{ m}^3}{1.08049 \text{ m}^3/\text{kg}} = 0.0139 \text{ kg}$$

Substituting, $Q_{\text{out}} = (0.0139 \text{ kg})(2654.6 - 1748.7) \text{ kJ/kg} = 12.58 \text{ kJ}$

The volume and the mass of the air in the room are $V = 4 \times 4 \times 5 = 80 \text{ m}^3$ and

$$m_{\text{air}} = \frac{P_1 V_1}{RT_1} = \frac{(100 \text{ kPa})(80 \text{ m}^3)}{(0.2870 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(283 \text{ K})} = 98.5 \text{ kg}$$

The amount of fan work done in 30 min is

$$W_{\text{fan,in}} = \dot{W}_{\text{fan,in}} \Delta t = (0.120 \text{ kJ/s})(30 \times 60 \text{ s}) = 216 \text{ kJ}$$

We now take the air in the room as the system. The energy balance for this closed system is expressed as

$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

$$Q_{\text{in}} + W_{\text{fan,in}} - W_{\text{b,out}} = \Delta U$$

$$Q_{\text{in}} + W_{\text{fan,in}} = \Delta H \cong mc_p(T_2 - T_1)$$

since the boundary work and ΔU combine into ΔH for a constant pressure expansion or compression process. It can also be expressed as

$$(\dot{Q}_{\text{in}} + \dot{W}_{\text{fan,in}}) \Delta t = mc_{p,\text{avg}}(T_2 - T_1)$$

Substituting, $(12.58 \text{ kJ}) + (216 \text{ kJ}) = (98.5 \text{ kg})(1.005 \text{ kJ/kg}^\circ\text{C})(T_2 - 10)^\circ\text{C}$

which yields $T_2 = 12.3^\circ\text{C}$

Therefore, the air temperature in the room rises from 10°C to 12.3°C in 30 min.

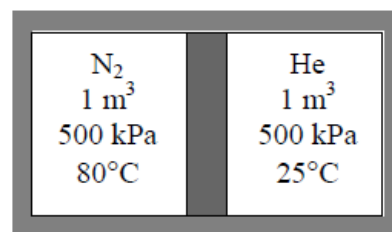
4-138 Consider a well-insulated horizontal rigid cylinder that is divided into two compartments by a piston that is free to move but does not allow either gas to leak into the other side. Initially, one side of the piston contains 1 m^3 of N_2 gas at 500 kPa and 80°C while the other side contains 1 m^3 of He gas at 500 kPa and 25°C . Now thermal equilibrium is established in the cylinder as a result of heat transfer through the piston. Using constant specific heats at room temperature, determine the final equilibrium temperature in the cylinder. What would your answer be if the piston were not free to move?

Properties The gas constants and the constant volume specific heats are $R = 0.2968 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K}$ is $c_v = 0.743 \text{ kJ/kg}\cdot^\circ\text{C}$ for N_2 , and $R = 2.0769 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K}$ is $c_v = 3.1156 \text{ kJ/kg}\cdot^\circ\text{C}$ for He (Tables A-1 and A-2)

Analysis The mass of each gas in the cylinder is

$$m_{\text{N}_2} = \left(\frac{P_1 V_1}{RT_1} \right)_{\text{N}_2} = \frac{(500 \text{ kPa})(1 \text{ m}^3)}{(0.2968 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(353 \text{ K})} = 4.77 \text{ kg}$$

$$m_{\text{He}} = \left(\frac{P_1 V_1}{RT_1} \right)_{\text{He}} = \frac{(500 \text{ kPa})(1 \text{ m}^3)}{(2.0769 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(298 \text{ K})} = 0.808 \text{ kg}$$



Taking the entire contents of the cylinder as our system, the 1st law relation can be written as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{\text{system}}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc. energies}}}$$

$$0 = \Delta U = (\Delta U)_{\text{N}_2} + (\Delta U)_{\text{He}}$$

$$0 = [mc_v(T_2 - T_1)]_{\text{N}_2} + [mc_v(T_2 - T_1)]_{\text{He}}$$

Substituting,

$$(4.77 \text{ kg})(0.743 \text{ kJ/kg}\cdot^\circ\text{C})(T_f - 80)^\circ\text{C} + (0.808 \text{ kg})(3.1156 \text{ kJ/kg}\cdot^\circ\text{C})(T_f - 25)^\circ\text{C} = 0$$

It gives $T_f = 57.2^\circ\text{C}$

where T_f is the final equilibrium temperature in the cylinder.

The answer would be the **same** if the piston were not free to move since it would effect only pressure, and not the specific heats.

4-139 Repeat Prob. 4-138 by assuming the piston is made of 5 kg of copper initially at the average temperature of the two gases on both sides. *Answer: 56°C*

Analysis The mass of each gas in the cylinder is

$$m_{N_2} = \left(\frac{P_1 V_1}{RT_1} \right)_{N_2} = \frac{(500 \text{ kPa})(1 \text{ m}^3)}{(0.2968 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(353 \text{ K})} = 4.77 \text{ kg}$$

$$m_{He} = \left(\frac{P_1 V_1}{RT_1} \right)_{He} = \frac{(500 \text{ kPa})(1 \text{ m}^3)}{(2.0769 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(353 \text{ K})} = 0.808 \text{ kg}$$

Taking the entire contents of the cylinder as our system, the 1st law relation can be written as

$$\underbrace{E_{in} - E_{out}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{system}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$0 = \Delta U = (\Delta U)_{N_2} + (\Delta U)_{He} + (\Delta U)_{Cu}$$

$$0 = [mc_v(T_2 - T_1)]_{N_2} + [mc_v(T_2 - T_1)]_{He} + [mc(T_2 - T_1)]_{Cu}$$

where

$$T_{1, Cu} = (80 + 25) / 2 = 52.5^\circ\text{C}$$

Substituting,

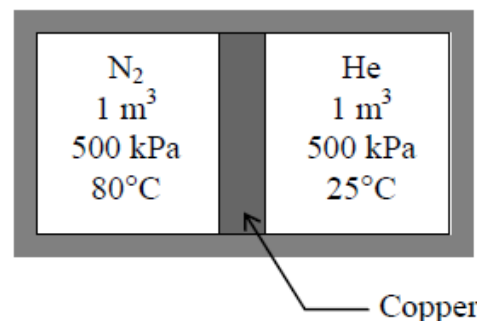
$$(4.77 \text{ kg})(0.743 \text{ kJ/kg} \cdot ^\circ\text{C})(T_f - 80)^\circ\text{C} + (0.808 \text{ kg})(3.1156 \text{ kJ/kg} \cdot ^\circ\text{C})(T_f - 25)^\circ\text{C} \\ + (5.0 \text{ kg})(0.386 \text{ kJ/kg} \cdot ^\circ\text{C})(T_f - 52.5)^\circ\text{C} = 0$$

It gives

$$T_f = 56.0^\circ\text{C}$$

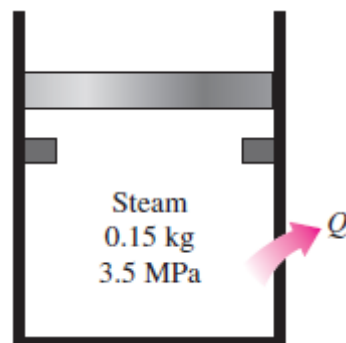
where T_f is the final equilibrium temperature in the cylinder.

The answer would be the **same** if the piston were not free to move since it would effect only pressure, and not the specific heats.



4-142 A vertical 12-cm diameter piston–cylinder device contains an ideal gas at the ambient conditions of 1 bar and 24°C. Initially, the inner face of the piston is 20 cm from the base of the cylinder. Now an external shaft connected to the piston exerts a force corresponding to a boundary work input of 0.1 kJ. The temperature of the gas remains constant during the process. Determine (a) the amount of heat transfer, (b) the final pressure in the cylinder, and (c) the distance that the piston is displaced.

4-143 A piston–cylinder device initially contains 0.15-kg steam at 3.5 MPa, superheated by 5°C. Now the steam loses heat to the surroundings and the piston moves down, hitting a set of stops at which point the cylinder contains saturated liquid water. The cooling continues until the cylinder contains water at 200°C. Determine (a) the final pressure and the quality (if mixture), (b) the boundary work, (c) the amount of heat transfer when the piston first hits the stops, (d) and the total heat transfer.



Chapter 4

End of the Problems